

Poisson Prob Distribution

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(Discrete Prob distⁿ)

Defⁿ: A discrete r.v. X is said to follow a Poisson prob distribution with parameter λ ($\lambda > 0$), if its prob mass function is given as

$$P(X=x) = p(x) = \frac{e^{-\lambda} \lambda^x}{x!}; x=0,1,2,\dots,\infty$$

Remark: I) $X \sim P(\lambda)$; $\lambda > 0$

II) $p(x)$ is pmf, i.e. $p(x) \geq 0 \forall x=0,1,\dots,\infty$

$$\sum_{x=0}^{\infty} p(x) = \sum_{x=0}^{\infty} \frac{e^{-\lambda} \lambda^x}{x!} = e^{-\lambda} e^{\lambda} = e^0 = 1$$

Examples: I) No. of deaths from a disease (not in the form of an epidemic) such as heart attack, cancer or due to snake bite.

II) No. of suicides reported in a particular city

III) No. of defective materials in a packing manufactured by a good concern (IV) No. of faulty blades in a packet of 100.

V) No. of air accidents in some unit of time

VI) No. of printing mistakes at each page of the book

VII) No. of telephone calls received at a particular telephone exchange in some units of time or connections to wrong numbers in a telephone exchange

VIII) No. of cars passing a crossing per minute during the busy hours of the day

IX) The No. of fragments received by a surface area 'A' from a fragment atom bomb

X) The emission of radioactive (alpha) particles.

Theorem: Poisson distⁿ is the limiting form of binomial prob distⁿ

or

$$B(n, p) \longrightarrow P(\lambda)$$

Assumptions: I) n (no. of finite bernoulli trials) $\longrightarrow \infty$ (For numerical when $n > 30$) (too large)

II) Prob of success $p \longrightarrow 0$ (too small) such that $np = \lambda$ ($\lambda > 0$) (finite), gives $p = \frac{\lambda}{n}$

Proof

$$X \sim B(n, p)$$

$$p(x) = \binom{n}{x} p^x (1-p)^{n-x}; \quad x = 0, 1, \dots, n$$

$$= \frac{n!}{x! (n-x)!} \left(\frac{p}{1-p}\right)^x (1-p)^n$$

$$= \frac{n(n-1) \dots (n-x+1) \cancel{1!}}{x! \cancel{1!}} \frac{\left(\frac{\lambda}{n}\right)^x}{\left(1 - \frac{\lambda}{n}\right)^x} \left(1 - \frac{\lambda}{n}\right)^n$$

$$= \frac{\cancel{n!} \cdot 1 \cdot \left(1 - \frac{1}{n}\right) \left(1 - \frac{2}{n}\right) \dots \left(1 - \frac{x-1}{n}\right) \left(\frac{\lambda}{n}\right)^x \left(1 - \frac{\lambda}{n}\right)^n}{x! \left(1 - \frac{\lambda}{n}\right)^x}$$

taking limit $n \rightarrow \infty$, we have

$$\lim_{n \rightarrow \infty} B(n, p) \Rightarrow p(x) = \frac{\lambda^x e^{-\lambda}}{x!}, \quad x = 0, 1, 2, \dots, \infty$$

Hence, it is proved

As we know $\lim_{n \rightarrow \infty} \left(1 - \frac{\lambda}{n}\right)^n = e^{-\lambda}$
 $\lim_{n \rightarrow \infty} \left(1 - \frac{\lambda}{n}\right)^x = 1$

Moment-Generating Function of $P(\lambda)$

$$X \sim P(\lambda) \quad ; \quad p(x) = \frac{e^{-\lambda} \lambda^x}{x!}; \quad x = 0, 1, \dots, \infty$$

$$\begin{aligned} \text{MGF } M_X(t) &= E e^{tx} = \sum_{x=0}^{\infty} e^{tx} \cdot p(x) = \sum_{x=0}^{\infty} e^{tx} \frac{e^{-\lambda} \lambda^x}{x!} \\ &= e^{-\lambda} \sum_{x=0}^{\infty} \frac{(\lambda e^t)^x}{x!} = e^{-\lambda} e^{\lambda e^t} = e^{\lambda(e^t - 1)} \end{aligned}$$

$$\boxed{M_X(t) = e^{\lambda(e^t - 1)}}$$

$$\text{Mean } \mu_1' = \left. \frac{\partial M_X(t)}{\partial t} \right|_{t=0}; \quad \frac{\partial M_X(t)}{\partial t} = e^{\lambda(e^t - 1)} \cdot \lambda e^t$$

$$\therefore \text{Mean } (\mu_1') = \left. \frac{\partial M_X(t)}{\partial t} \right|_{t=0} = e^{\lambda(e^0 - 1)} \lambda e^0 = \lambda$$

$$\boxed{\text{Mean} = \lambda}$$

$$\mu_2' = \left. \frac{\partial^2 M_X(t)}{\partial t^2} \right|_{t=0}; \quad \frac{\partial^2 M_X(t)}{\partial t^2} = e^{\lambda(e^t - 1)} (\lambda e^t)^2 + e^{\lambda(e^t - 1)} \lambda e^t$$

$$\therefore \mu_2' = \frac{\partial^2 M_X(t)}{\partial t^2} \bigg|_{t=0} = \lambda^2 + \lambda$$

$$\therefore \text{variance } \mu_2 = \mu_2' - \mu_1'^2 \\ = \lambda^2 + \lambda - \lambda^2 = \lambda$$

$$\boxed{\text{Variance} = \lambda}, \text{ Hence, in } P(\lambda)$$

$$\boxed{\text{Mean} = \text{Variance} = \lambda}$$

Additive Property: If $X_1 \sim P(\lambda_1)$ and $X_2 \sim P(\lambda_2)$
 X_1 and X_2 are indep. r.v.s.

$$M_{X_1}(t) = e^{\lambda_1(e^t - 1)}, \quad M_{X_2}(t) = e^{\lambda_2(e^t - 1)}$$

$$M_{X_1+X_2}(t) = M_{X_1}(t) \cdot M_{X_2}(t) = e^{\lambda_1(e^t - 1)} \cdot e^{\lambda_2(e^t - 1)} \\ = e^{(\lambda_1 + \lambda_2)(e^t - 1)}; \text{ which the m.g.f. of } P(\lambda_1 + \lambda_2)$$

Hence, $X_1 + X_2 \sim P(\lambda_1 + \lambda_2)$; holds the additive property.

Ex. A car hire firm has two cars, which it hires out day by day. The number of demands for a car on each day is distributed as a Poisson distⁿ with mean 1.5. calculate the proportions of days on which (i) neither car is used, and (ii) the proportions of days on which some demand is refused.

X : No. of demands $X \sim P(1.5); \lambda = 1.5$

$$p(x) = \frac{e^{-1.5} (1.5)^x}{x!}; x = 0, 1, 2, \dots$$

$$i) P(X=0) = p(0) = e^{-1.5} = 0.2231$$

$$ii) P(X > 2) = 1 - P(X \leq 2) = 1 - (p(0) + p(1) + p(2)) \\ = 1 - e^{-1.5} \left(1 + 1.5 + \frac{(1.5)^2}{2} \right) = 0.19126.$$

Ex. A manufacturer of letter pins knows that 5% of his product is defective. If he sells letter pins in boxes of 100 and guarantees that not more than 10 pins will be defective, what is the approximate probability that a box will fail to meet the guaranteed quality.

Soln: $n = 100$, $p = 5\% = 0.05$, X : No. of defectives
 p (small) and n is large, we apply Poisson approximation. $\lambda = np = 100 \times 0.05 = 5$

$$p(x) = \frac{e^{-\lambda} \lambda^x}{x!}; \quad x = 0, 1, \dots, \infty$$

Box will fail to meet the guaranteed quality if $X > 10$
 hence, $P(X > 10) = 1 - P(X \leq 10) = 1 - \sum_{x=0}^{10} \frac{e^{-5} (5)^x}{x!}$

$$P(X > 10) = 1 - e^{-5} \sum_{x=0}^{10} \frac{5^x}{x!}$$

Ex. A manufacturer, who produces medicine bottles, finds that 0.1% of the bottles are defective. The bottles are packed in boxes containing 500 bottles. A drug manufacturer buys 100 boxes from the producer of bottles, find how many boxes will contain 1) no defective 2) at least two defectives.

Here $p = 0.1\% = 0.001$ (small)

$n = 500$ (large), $N = 100$

$\therefore \lambda = np = 500 \times 0.001 = 0.5$

We apply Poisson approximation.

$$1) \quad f(0) = N p(0) = 100 \times \frac{e^{-0.5} (0.5)^0}{0!} = 100 \times e^{-0.5} \\ = 100 \times 0.6065 \approx 61$$

$$2) \quad f(x \geq 2) = NP(X \geq 2) = N [1 - P(X \leq 1)] \\ = N [1 - p(0) - p(1)] = 100 [1 - 0.6065 - 0.5 \times 0.6065] \approx 9$$

Normal Prob Distribution

(Cont. prob distⁿ defined for cont. r.v.)

Defⁿ: A continuous r.v. X is said to follow a normal prob distⁿ with mean μ and variance σ^2 , if its prob. density function is given as

$$f(x, \mu, \sigma^2) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\frac{(x-\mu)^2}{\sigma^2}}; \quad -\infty < x < \infty$$

$$\quad \quad \quad -\infty < \mu < \infty$$

$$\quad \quad \quad \sigma^2 > 0$$

Where μ and σ^2 are known as the parameters of normal prob distⁿ.

Remark: i) $X \sim N(\mu, \sigma^2)$

ii) If $X \sim N(\mu, \sigma^2)$ then $Z = \frac{X-\mu}{\sigma}$ known as standard normal variate with mean 0 and variance 1.

i.e. $Z \sim N(0, 1)$; its pdf is given as

$$f(z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}z^2}; \quad -\infty < z < \infty$$

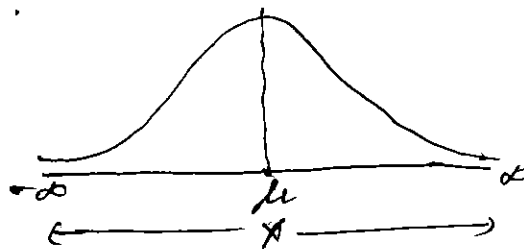
The corresponding distⁿ $\phi(z)$ is given as

$$\phi(z) = P(Z \leq z) = \int_{-\infty}^z f(z) dz$$

→ Chief characteristics of the normal distⁿ and Normal prob curve.

If $X \sim N(\mu, \sigma^2)$; $f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\frac{(x-\mu)^2}{\sigma^2}}, -\infty < x < \infty$

i) The normal prob curve is bell-shaped and symmetrical about the line $x = \mu$.



ii) Mean, median and mode of the distⁿ coincides.

i.e. Mean = Median = Mode = μ

iii) As x increases numerically, $f(x)$ decreases rapidly, the maximum prob occurring at the point $x = \mu$: $[f(x)]_{\max} = \frac{1}{\sigma\sqrt{2\pi}}$.

iv) $\beta_1 = 0$ and $\beta_2 = 3$

v) $\mu_{2r+1} = 0, (r = 0, 1, 2, \dots)$ and $\mu_{2r} = 1 \cdot 3 \cdot 5 \cdot \dots \cdot (2r-1)\sigma^2, (r = 0, 1, 2, \dots)$

- VI) Since $f(x)$ being the probability, can never be negative, no portion of the curve lies below the x axis.
- VII) Linear combination of independent normal variates is also a normal variate.

i.e. $X_1 \sim N(\mu_1, \sigma_1^2), X_2 \sim N(\mu_2, \sigma_2^2) \dots X_k \sim N(\mu_k, \sigma_k^2)$ are k indep. normal variates, then

$$a_1 X_1 + a_2 X_2 + \dots + a_k X_k \sim N\left(\sum_{i=1}^k a_i \mu_i, \sum_{i=1}^k a_i^2 \sigma_i^2\right)$$

Ex. $X_1 \sim N(1, 2), X_2 \sim N(-2, 3)$

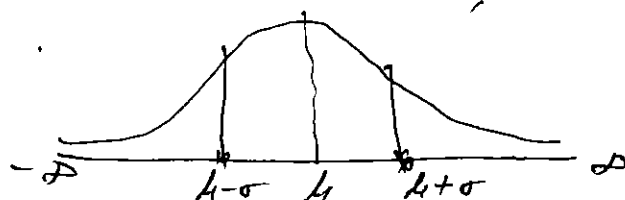
then $2X_1 - X_2 \sim N(2 \times 1 - (-2), 4 \times 2 + 1 \times 3)$
 $\sim N(4, 7)$

Similarly $2X_1 + X_2 \sim N(0, 7)$

- VIII) x -axis is an asymptote to the normal prob curve.

- IX) The points of inflexion of the curve are $x = \mu \pm \sigma$,

$$f(x) \Big|_{x=\mu \pm \sigma} = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}}$$



- X) Mean deviation about mean $= \sqrt{\frac{2}{\pi}} \cdot \sigma \approx \frac{4}{5} \sigma$ (approx.)

- (XI) Quartiles are given by $Q_1 = \mu - 0.6745\sigma, Q_3 = \mu + 0.6745\sigma$

XII) A.D. $= \frac{Q_3 - Q_1}{2} \approx \frac{2}{2} \sigma$ (approx)

A.D. : M.D. : S.D. :: $\frac{2}{\sigma} \sigma : \frac{4}{5} \sigma : \sigma :: \frac{2}{2} : \frac{4}{5} : 1$

$\Rightarrow$ A.D. : M.D. : S.D. :: 10 : 12 : 15

- XIII) Area Property: $P(\mu - \sigma < X < \mu + \sigma) = 0.6826$

$$P(\mu - 2\sigma < X < \mu + 2\sigma) = 0.9544$$

$$P(\mu - 3\sigma < X < \mu + 3\sigma) = 0.9974$$

- XIV) If X and Y are independent standard normal variates, then it can be easily proved that $U = X + Y$ and $V = X - Y$ are independently distributed, $U \sim N(0, 2)$ and $V \sim N(0, 2)$.

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[illegible]

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MGF of Normal distribution $X \sim N(\mu, \sigma^2)$

$$M_X(t) = E e^{tX} = \int_{-\infty}^{\infty} e^{tx} f(x) dx = \frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{tx} e^{-\frac{1}{2\sigma^2}(x-\mu)^2} dx$$

$$Z = \frac{x-\mu}{\sigma}$$

$$\sigma dz = dx$$

$$\therefore M_X(t) = \frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{t(\mu+\sigma z)} e^{-\frac{z^2}{2}} \sigma dz$$

$$= e^{\mu t} \cdot \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{1}{2}(z^2 - 2t\sigma z)} dz$$

$$= e^{\mu t} \cdot \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{1}{2}\{ (z-\sigma t)^2 - \sigma^2 t^2 \}} dz$$

$$= e^{\mu t + \frac{1}{2}\sigma^2 t^2} \cdot \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{1}{2}(z-\sigma t)^2} dz$$

$$= e^{\mu t + \frac{1}{2}\sigma^2 t^2} \cdot \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{u^2}{2}} du \quad ; \quad u = z - \sigma t, \quad du = dz$$

Hence

$$\boxed{M_X(t) = e^{\mu t + \frac{1}{2}\sigma^2 t^2}}$$

$$\int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{u^2}{2}} du = 1$$

pdf of SNV.

$$Z = \frac{X-\mu}{\sigma} \sim N(0, 1)$$

$$M_Z(t) = e^{\frac{1}{2}t^2}$$

How to calculate Normal prob?

Area property: 1) $P(\mu - \sigma < X < \mu + \sigma) = 0.6826$

$$= \int_{\mu-\sigma}^{\mu+\sigma} f(x) dx = \int_{\mu-\sigma}^{\mu+\sigma} \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\frac{(x-\mu)^2}{\sigma^2}} dx$$

$$\frac{x-\mu}{\sigma} = Z \Rightarrow x = \mu + \sigma Z$$

$$dx = \sigma dz$$

$$= \int_{-1}^1 \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}z^2} \sigma dz = \int_{-1}^1 \phi(z) dz = P(-1 < Z < 1)$$

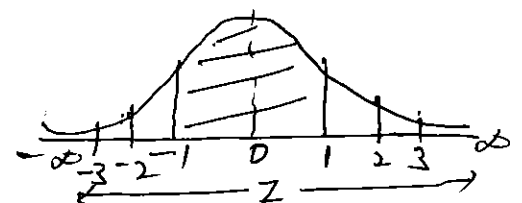
$$= 2P(0 < Z < 1) = 2 \times 0.3413 = 0.6826$$

Similarly $P(\mu - 2\sigma < X < \mu + 2\sigma)$

$$= P(-2 < Z < 2) = 0.9544$$

$$P(\mu - 3\sigma < X < \mu + 3\sigma)$$

$$= P(-3 < Z < 3) = 0.9974$$



Ex. $X \sim N(12, 16)$; $\mu=12, \sigma^2=16, \sigma=4$

(a) Find (i) $P(X > 20)$ (ii) $P(X \leq 20)$ and (iii) $P(0 \leq X \leq 12)$

(b) Find x' , when $P(X > x') = 0.24$

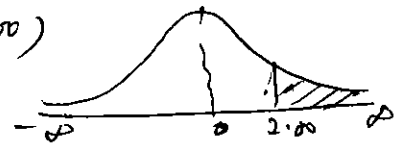
(c) Find x'_0 and x'_1 , when $P(x'_0 < X < x'_1) = 0.50$ and $P(X > x'_1) = 0.25$

Table is given for standard normal variate $Z \sim N(0, 1)$

$$(a) (i) P(X > 20) = P\left(\frac{X-\mu}{\sigma} > \frac{20-\mu}{\sigma}\right) = P\left(Z > \frac{20-12}{4}\right)$$

$$= P(Z > 2) = 0.5 - P(0 \leq Z \leq 2.00)$$

$$= 0.5 - 0.4772 = 0.0228$$



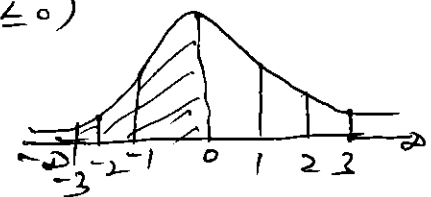
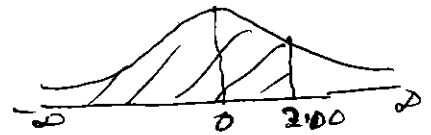
$$(ii) P(X \leq 20) = P\left(\frac{X-\mu}{\sigma} \leq \frac{20-12}{4}\right) = P(Z \leq 2)$$

$$= 0.5 + P(0 \leq Z \leq 2.00) = 0.5 + 0.4772 = 0.9772$$

$$(iii) P(0 \leq X \leq 12) = P\left(\frac{0-\mu}{\sigma} \leq \frac{X-\mu}{\sigma} \leq \frac{12-\mu}{\sigma}\right)$$

$$= P\left(\frac{0-12}{4} \leq Z \leq \frac{12-12}{4}\right) = P(-3 \leq Z \leq 0)$$

$$= P(0 \leq Z \leq 3.00) = 0.4987$$



(b) Find x' when $P(X > x') = 0.24$

$$P\left(\frac{X-\mu}{\sigma} > \frac{x'-\mu}{\sigma}\right) = 0.24 \Rightarrow P\left(Z > \frac{x'-12}{4}\right) = 0.24$$

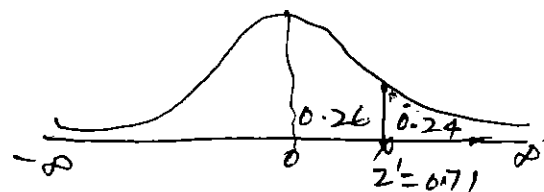
$$= P(Z > z') = 0.24$$

From table $z' = 0.71$

$$= \frac{x'-12}{4} = 0.71$$

$$x' = 12 + 4 \times 0.71$$

$$= 12 + 2.84 = 14.84$$

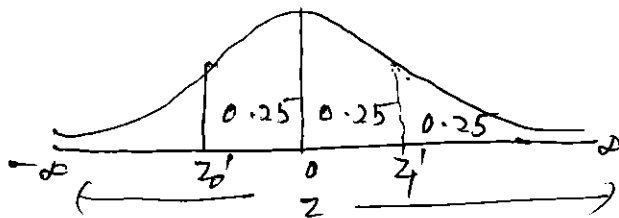


(c) Find x'_0 and x'_1 , when $P(x'_0 < X < x'_1) = 0.50$ and $P(X > x'_1) = 0.25$

$$P\left(\frac{x'_0-12}{4} < \frac{X-\mu}{\sigma} < \frac{x'_1-12}{4}\right) = 0.50 \text{ and } P\left(\frac{X-\mu}{\sigma} > \frac{x'_1-12}{4}\right) = 0.25$$

$$\Rightarrow P(z'_0 < Z < z'_1) = 0.50 \text{ and } P(Z > z'_1) = 0.25$$

Now we have to locate the points z'_0 and z'_1 so that above conditions are satisfied.



From table $z_1' = 0.67$, $z_0' = -0.67$

Hence $z_0' = -0.67$

$$\frac{x_0' - 12}{4} = -0.67 \Rightarrow x_0' = 12 - 4 \times 0.67 = 12 - 2.68 = \underline{9.32}$$

$$z_1' = 0.67$$

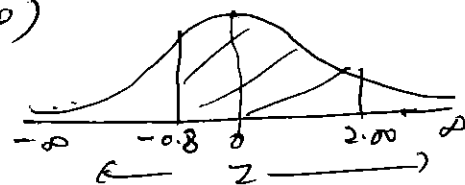
$$\frac{x_1' - 12}{4} = 0.67 \Rightarrow x_1' = 12 + 4 \times 0.67 = 12 + 2.68 = \underline{14.68}$$

Ex. $X \sim N(30, 25)$, Find

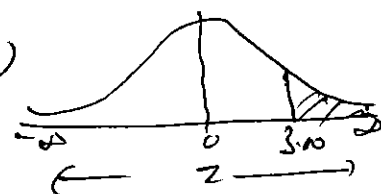
i) $P(26 \leq X \leq 40)$ (ii) $P(X > 45)$ (iii) $P(|X - 30| > 5)$

Given $\mu = 30$, $\sigma^2 = 25$, $\sigma = 5$

$$\begin{aligned} \text{i) } P\left(\frac{26-30}{5} \leq \frac{X-\mu}{\sigma} \leq \frac{40-30}{5}\right) &= P(-0.8 \leq Z \leq 2.00) \\ &= P(-0.8 \leq Z \leq 0) + P(0 \leq Z \leq 2.00) \\ &= 0.2881 + 0.4772 = \underline{0.7653} \end{aligned}$$



$$\begin{aligned} \text{ii) } P(X > 45) &= P\left(\frac{X-\mu}{\sigma} > \frac{45-30}{5}\right) \\ &= P(Z > 3.00) = 0.5 - P(0 \leq Z \leq 3.00) \\ &= 0.5 - 0.4987 = \underline{0.0013} \end{aligned}$$



$$\begin{aligned} \text{iii) } P(|X - 30| > 5) &= 1 - P(|X - 30| \leq 5) \\ &= 1 - P(-5 \leq (X - 30) \leq 5) \\ &= 1 - P\left(-\frac{25-30}{5} \leq \frac{X-\mu}{\sigma} \leq \frac{35-30}{5}\right) \\ &= 1 - P(-1 \leq Z \leq 1) = 1 - 2P(0 \leq Z \leq 1.00) = 1 - 2 \times 0.3413 \\ &= 1 - 0.6826 = \underline{0.3174} \end{aligned}$$

Ex. In a normally distributed population, 10.03% items are under 25 and 89.97% items are under 70. What are the mean and S.d. of the distribution.

Given $X \sim N(\mu, \sigma^2)$

$$P(X < 25) = 0.1003 \quad \text{and} \quad P(X < 70) = 0.8997$$

$$P\left(\frac{X-\mu}{\sigma} < \frac{25-\mu}{\sigma}\right) = 0.1003 \quad \text{and} \quad P\left(\frac{X-\mu}{\sigma} < \frac{70-\mu}{\sigma}\right) = 0.8997$$

$$P(Z < Z_0) = 0.1003 \quad \text{and} \quad P(Z < Z_1) = 0.8997$$

$$Z_0 = -1.28$$

$$\frac{25-\mu}{\sigma} = -1.28$$

$$25 = \mu - 1.28\sigma$$

$$\Rightarrow 25 - \mu = -1.28\sigma \quad \text{--- (A)}$$

again $Z_1 = 1.28$

$$\frac{70-\mu}{\sigma} = 1.28 \Rightarrow 70 - \mu = 1.28\sigma \quad \text{--- (B)}$$

Solving (A) + (B) we get $\mu = 47.5$ and $\sigma = 17.578$

